

Low-Power Multi-Node IoT Sensor Network with Energy Analysis

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Abstract

This project presents the design, implementation, and evaluation of a low-power multi-node Internet of Things (IoT) sensor network. The system consists of two battery-powered sensor nodes and a central gateway. Node 1 measures temperature and humidity using a DHT22 sensor, while Node 2 measures ambient light intensity using a Light Dependent Resistor (LDR). Both nodes communicate wirelessly with an ESP8266 gateway using nRF24L01 radio modules. The gateway uploads the collected data to the ThingSpeak cloud platform for real-time visualization and storage. Special emphasis is placed on low-power operation through duty cycling and deep sleep modes. Energy consumption is experimentally measured using a digital multimeter, and battery lifetime as well as the number of operational cycles are estimated. Results demonstrate that duty cycling significantly extends battery lifetime compared to always-active operation.

1. Introduction

The rapid growth of IoT applications has increased the demand for energy-efficient sensor networks, particularly in scenarios where nodes are powered by batteries and deployed for long durations. Continuous operation of sensors and wireless communication modules leads to rapid battery depletion, making low-power design a critical requirement.

This project aims to design a practical IoT system that balances reliable sensing and communication with low energy consumption. By implementing duty-cycled operation, sleep modes, and a gateway-based architecture, the system demonstrates how battery-powered sensor nodes can operate for extended periods while still providing real-time data to the cloud.

2. System Overview

The developed system follows a star network topology consisting of: - Two sensor nodes (Arduino Nano-based) - One gateway node (ESP8266-based) - A cloud platform (ThingSpeak)

Each sensor node periodically wakes up, reads its sensor, transmits the data wirelessly to the gateway, and then returns to deep sleep. The ESP8266 gateway receives data from all nodes and uploads it to the cloud for visualization and logging.

3. Hardware Architecture

3.1 Node 1: Temperature and Humidity Sensor Node

Node 1 is responsible for environmental monitoring and includes the following components: - Arduino Nano (ATmega328P) - DHT22 temperature and humidity sensor - nRF24L01 2.4 GHz RF transceiver - Status LED for active-mode indication - Two 3.7 V Li-ion batteries connected in series (7.4 V total)

3.2 Node 2: Light Intensity Sensor Node

Node 2 measures ambient light intensity and consists of: - Arduino Nano (ATmega328P) - LDR with a voltage divider circuit - nRF24L01 RF transceiver - Status LED for active-mode indication - Two 3.7 V Li-ion batteries connected in series (7.4 V total)

3.3 Gateway Node

The gateway node aggregates data from both sensor nodes and connects the system to the internet. It consists of: - ESP8266 microcontroller - nRF24L01 RF transceiver - Wi-Fi connectivity

The gateway is powered continuously and is not energy constrained, allowing it to handle cloud communication.

4. Software Architecture

4.1 Node Operation Flow

Each sensor node follows the same operational sequence:

1. Wake up from deep sleep
2. Turn on status LED (active mode)
3. Read sensor data
4. Transmit data via nRF24L01
5. Power down radio and peripherals
6. Turn off status LED
7. Enter deep sleep

This duty-cycled approach minimizes the average current consumption.

4.2 Low-Power Techniques

The following low-power techniques were implemented: - Deep sleep using LowPower.powerDown() on Arduino Nano - Powering down the RF transceiver when not transmitting - Short active periods (2–3 seconds) - Status LED disabled during sleep

5. Cloud Integration

Link: <https://thingspeak.mathworks.com/channels/3212137>

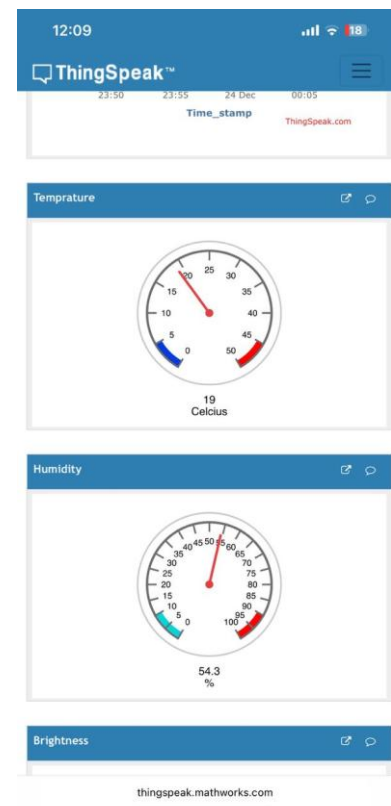
5.1 ThingSpeak Platform

The ESP8266 gateway uploads sensor data to the ThingSpeak cloud platform. The cloud dashboard provides: - Real-time numerical displays - Gauge-based visualization for temperature and humidity - Brightness visualization for LDR readings - Time-stamped data logging

5.2 Data Flow

Sensor Nodes → nRF24L01 → ESP8266 Gateway → Wi-Fi → ThingSpeak Cloud

This architecture keeps sensor nodes offline and low-power, while only the gateway handles internet communication.

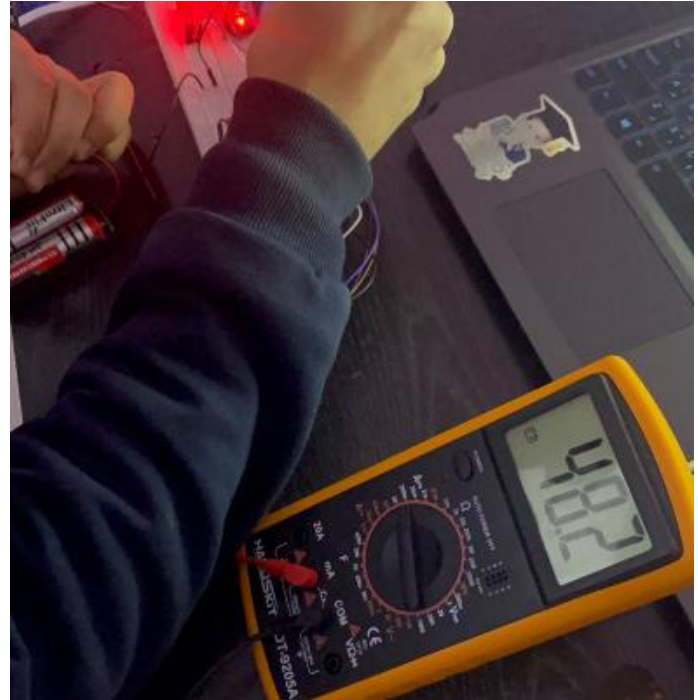


6. Experimental Setup for Energy Measurement

Energy consumption was measured experimentally using a digital multimeter (AVO meter). The meter was connected in series with the battery supply to measure current consumption.

Measurements were taken for: - Active mode (sensor reading and data transmission) - Sleep mode (deep sleep with peripherals powered down)

All measurements were performed while the nodes were powered by the actual Li-ion battery packs.



7. Measured Energy Consumption

7.1 Node 1 (DHT22)

- Active current: 48 mA
- Sleep current: 5.5 mA
- Active time: 3 s
- Sleep time: 16 s
- Total cycle time: 19 s

7.2 Node 2 (LDR)

- Active current: 24 mA
- Sleep current: 8.3 mA
- Active time: 2 s
- Sleep time: 24 s

8. Average Current Calculation

The average current was calculated using duty-cycle analysis:

$$I_{avg} = \frac{I_{active}T_{active} + I_{sleep}T_{sleep}}{T_{cycle}}$$

Results:

- Node 1: **12.21 mA**
- Node 2: **9.51 mA**

9. Comparison with Always-Active Operation

Node	Mode	Lifetime
Node 1	Always Active	~5.9 days
Node 1	Duty-Cycled	~23.2 days
Node 2	Always Active	~11.8 days
Node 2	Duty-Cycled	~29.8 days

No. cycles comparison:

Node	Cycles (Duty-Cycled)	Cycles (Always Active)
Node 1	~105,000	~22,000

Node	Cycles (Duty-Cycled)	Cycles (Always Active)
Node 2	~99,000	~41,000

10. Limitations

- Arduino Nano is not optimized for ultra-low-power sleep
- Sleep current remains in the milliampere range
- No hardware power gating for the microcontroller
- Battery voltage monitoring was not implemented

11. Conclusion

This project successfully demonstrates a low-power multi-node IoT sensor network with real-time cloud integration. Through duty cycling and sleep modes, battery lifetime was extended by several times compared to always-active operation. Experimental energy measurements validated the effectiveness of the design. The system provides a scalable and energy-aware foundation for long-term IoT monitoring applications.